Travis Courtney

Associate Professor, University of Puerto Rico Mayagüez
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Appointments and professional experience:

2024–present **Associate Professor**, University of Puerto Rico Mayagüez
2021–2024 **Assistant Professor**, University of Puerto Rico Mayagüez
2019–2021 **Postdoctoral Scholar**, Scripps Institution of Oceanography
2015–2019 **National Science Foundation Graduate Research Fellow**, Scripps Institution of Oceanography
2013–2014 **Visiting Scientist**, University of North Carolina at Chapel Hill
2013–2014 **Laboratory Technician**, Northeastern University
2013 **Education Intern**, North Carolina Aquarium at Pine Knoll Shores
2010–2013 **Undergraduate Research Assistant**, University of North Carolina at Chapel Hill

Education:

2019 **Ph.D. Oceanography**, Scripps Institution of Oceanography
Dissertation: "Quantifying the rates and drivers of coral and coral reef calcification in the Anthropocene"
Research Advisor: Dr. Andreas J. Andersson

2013 **B.S. Geological Sciences + B.S. Environmental Sciences**, University of North Carolina at Chapel Hill
Thesis: "Impact of atmospheric pCO_2 and seawater temperature on the stable isotopic composition ($\partial^{18}O$, $\partial^{13}O$) of echinoid calcite (*Echinometra viridis*)"
Research advisor: Dr. Justin B. Ries

Honors, scholarships, and awards:

2024 **Sloan Research Fellowship in Earth System Science**, Alfred P. Sloan Foundation
2020 **Chancellor's Dissertation Medal**, University of California San Diego
2019 **Graduate Student Excellence Research Award**, Scripps Institution of Oceanography
2019 **Travel Grant**, University of California San Diego Graduate Student Association
2018 **Graduate Student Excellence Travel Award**, Scripps Institution of Oceanography
2018 **Outstanding Mentor Award**, Scripps Graduate Peer Mentor Program
2016 **Outstanding Presentation Award**, Scripps Student Symposium
2016 **Scripps Fellowship**, Scripps Institution of Oceanography
2015 **Cody Fellowship**, Scripps Institution of Oceanography
2015 **Graduate Research Fellowship,** National Science Foundation
2013 **Graduated with Highest Honors and Distinction**, UNC at Chapel Hill
2013 **Environmental Excellence Award**, UNC at Chapel Hill Institute for the Environment
2013 **Carolina Research Scholar**, UNC at Chapel Hill Office of Undergraduate Research
2013 **1st Place Undergraduate Poster Award**, Anadarko Research Symposium
2012 **Roy L. Ingram Geology Fund Scholarship**, UNC at Chapel Hill Department of Geological Science
2012 **Judson Mead Geologic Field Station Scholarship**, Indiana University at Bloomington
2011 **Harrington Scholar**, UNC at Chapel Hill
2009 **Wrightsville Beach Longboard Association Scholarship**

Successful research grants and fellowships: $3,114,421

2024 Puerto Rico Sea Grant $210,001
"Optimizing the cultivation of Ulva seaweed in the Caribbean: developing the framework for larger-scale cultivation and inclusion into the Puerto Rico Blue Economy"
PI: Dr. Stacey Williams, **Dr. Travis Courtney**, Dr. Alexandra Gregory, Manuel Olmeda Saldaña

2024 Office of Naval Research $524,637
"Seafloor classification and elevation changes on coral reefs in La Parguera, Puerto Rico using an autonomous surface vehicle"
PI: **Dr. Travis Courtney**

2024	Office of Naval Research	~~$1,469,712~~
	This DEI Project was Terminated in Year 1 to Comply with Presidential Executive Order	*$148,829*
	"Inclusion and Belonging in Ocean Science (IBOS) Collaborative: The East Coast Division of the D-ENTERPRISE Initiative"	
	PI: **Dr. Travis Courtney,** Dr. Andre Amador, Dr. Johana Rotterová	
2024	Sloan Research Fellowship in Earth System Science	$75,000
	PI: **Dr. Travis Courtney**	
2024	United States Geological Survey	$319,999
	"Influence of coral reef restoration on reef-growth capacity, wave transformation, and shoreline protection in Puerto Rico"	
	PI: Dr. Andre Amador, **Dr. Travis Courtney**	
2023	Office of Naval Research	$208,025
	"Classification and structural complexity of marine substrates across spatial scales in Hawai'i using next-generation autonomous mapping vehicles"	
	PI: **Dr. Travis Courtney**	
2023	Schmidt Ocean Institute	$4,536
	"To support four students' participation in R/V Falkor (too)'s research expedition FKt230417"	
	PI: **Dr. Travis Courtney**	
2022	Office of Naval Research Subaward from Purdue University	$61,946
	"Blue Integrated Partnerships (BIP): Mentoring the Medici Scholars to Solve Tomorrow's Big Challenges"	
	PI: Dr. Juan José Cruz Motta, **Dr. Travis Courtney**	
2022	Puerto Rico Department of Natural and Environmental Resources	$1,324,698
	"Development and Implementation of a Water Quality Monitoring Project in Shallow Coral Reef Areas Around Puerto Rico, and the Implementation of a 'BCG' Model"	
	PI: Dr. Juan José Cruz Motta, Dr. Ernesto Weil, **Dr. Travis Courtney**	
2022	EcoEléctrica	$94,925
	"Quantifying net ecosystem metabolism and biogeochemical variability of seagrasses in Southwest Puerto Rico"	
	PI: **Dr. Travis Courtney,** Catherine Hernandez Rodriguez [MS student in lab of PI Courtney]	
2016	Shepard Foundation Student Fieldwork	$1,840
	"Seasonal variability of net coral reef calcification in Kāne'ohe Bay, Hawai'i"	
	PI: **Travis Courtney**	
2016	Shepard Foundation Student Fieldwork	$1,985
	"Impacts of widespread bleaching on net coral reef calcification in Kāne'ohe Bay, Hawai'i"	
	PI: **Travis Courtney**	
2015	National Science Foundation Graduate Research Fellowship Program	$138,000
	PI: **Travis Courtney**	

Peer-reviewed publications: Mentored Student in Biogeochemistry and Ecology Research Group

48. Toth LT and **Courtney TA**. 2026. Repeated coral bleaching events have eroded reef accretion potential in the Florida Keys. Frontiers in Marine Science 13:1878462. https://doi.org/10.3389/fmars.2026.1878462
47. Kekuewa SAH, Rintoul MS, **Courtney TA**, Pezner AK, Takeda T, Kayanne H, Mitarai S, Andersson AJ. 2026. Seasonal Reef-Scale Variability in Seawater CO_2 Chemistry in the Presence and Absence of Seaweed Cultivation in Onna, Okinawa, Japan. *Estuaries and Coasts* 49(168). https://doi.org/10.1007/s12237-026-01770-x
46. Cordero E, Sherman CE, Yadav P, Kumar Raju V, Townsend JE, **Courtney TA**. 2026. Coral Intel: a YOLOv8 deep learning framework for monitoring Caribbean corals. Intelligent Marine Technology and Systems 4(18). https://doi.org/10.1007/s44295-026-00107-5

45. Infante R, Centeno L, **Courtney TA**, Cruz Motta JJ, Armstrong RA. 2026. Quantifying Marine Surface Microplastics in La Parguera Natural Reserve, Puerto Rico. Microplastics 5(108). https://doi.org/10.3390/microplastics5020108
44. DeCarlo TM, Jasnos O, Strange A, Andersson A, Bautista VII AT, Bloomer S, Bolden IW, Bosman M, Brachert TC, Braz GB, Cardoso GO, Carricart-Ganivet JP, Carilli JE, Castillo KD, Cavole L, Chan S, Chen X, Chomitz B, Correge T, **Courtney TA**, Deigan M, D'Olivo JP, Dunbar R, Enochs IC, Falsarella L, Felis T, Gutierrez-Estrada G, Hedger B, Hu S, Jameson S, Jupiter S, Kench P, Kersting DK, Lin K, Liu Y-W, Lorigados CAB, Manzello DP, McCulloch MT, Mies M, Moura RL, Oberle F, Pereira N, Prouty N, Ramos RD, Ren H, Ryan E, Thompson DM, Toth LT, Vergotti MJ, Webster JM, Zinke J. 2026. CoralCache: a virtual coral core repository for transparent and reproducible annual growth rate analyses, Earth System Science Data 18, 3341–3353. https://doi.org/105194/essd-18-3341-2026
43. Kekuewa SAH, Pezner AK, **Courtney TA**, Wei Y, Rintoul MS, Chou W-C, Soong K, Andersson AJ. 2026. Present and future seawater CO_2 chemistry across multiple coral reef habitats and scales at Dongsha Atoll and Taiping Island in the South China Sea. Coral Reefs. https://doi.org/10.1007/s00338-026-02868-1
42. Toth LT, Borer ET, Burkepile DE, Dudney J, Lemoine NP, Renzi JJ, Smith KEL, **Courtney TA**, Goeking SA, Hammond WM, Hoover DL, MacFayden S, Osland MJ, Townsend JE, Fidler RY. 2026. Scenarios and strategies for future-proofing ecosystem management under climatic novelty. Conservation Biology. e70250. https://doi.org/10.1111/cobi.70250
41. Cornwall CE, Timmerman O, Andersson A, Berbece DM, Branson O, Comeau S, **Courtney TA**, Donner SD, Enochs I, Harvey BP, Isaak AL, Koch HE, Logan CA, Perry CT, Schoepf V, Toth LT, Webb A. 2026. Persistence of coral reef structures into the twenty-first century. Nature Reviews Earth & Environment. 2026 Feb 19:1-1. https://doi.org/10.1038/s43017-026-00764-4
40. **Courtney TA**, Ricaurte-Chica M, Bayo Torres AD, Chabrier-Alpi AM, Medina Martinez AG, Melendez Declet CV, Mejías-Rivera CL, Fontanez Diaz CF, Hernández Rodríguez C, Lebron-Maldanado CI, Diaz Jimenez E, Irizarry Torres I, Carballeira Martínez J, Brito-Vera MJ, Luquis Ramos I, Santos Echevarría I, Collazo Reyes JJ, Centeno Mejias LJ, Nieves-Ortíz MA, Olmeda Saldaña MF, Gordon M, Mendez Falcon V, Reyes Claudio VR, Garcia Fernandez A, Melendez Aguilar AR, Garcia Hernandez WE, Feliberty Ruiz A, Cruz-Motta JJ. Puerto Rico Coral Reef Monitoring Program Water Quality Data from 2023–2025. 2026. Scientific Data 13(150). https://doi.org/10.1038/s41597-025-06468-6
39. Luquis Ramos IM, Williams SM, Cruz-Motta JJ, **Courtney TA**. 2026. Modeling bioerosion rates by *Diadema antillarum* populations to improve coral reef restoration in the Caribbean. Restoration Ecology. https://doi.org/10.1111/rec.70302
38. Rodríguez Hernández C, Reyes JC, Perez DI, Gonzalez-Corredor J, Krause JR, Campbell JE, Fourqurean JW, Cruz Motta JJ, **Courtney TA**. 2026. Influence of Environmental Factors on pH Buffering, Productivity, and Blue Carbon Storage in Contrasting *Thalassia testudinum* Seagrass Meadows in La Parguera, Puerto Rico. Estuaries and Coasts 49(40). https://doi.org/10.1007/s12237-025-01652-8
37. Mejias-Rivera CL, Ricaurte-Chica M, Carballeira-Martínez J, Hernández WJ, Armstrong RA, Cruz-Motta JJ, **Courtney TA**. 2026. Assessment of Coral Reef Eutrophication Thresholds Across Puerto Rico Using In-situ and Satellite-derived Chlorophyll-*a*. Estuaries and Coasts 49(26). https://doi.org/10.1007/s12237-025-01644-8
36. Martinez Ortiz J, Perez Perez J, Armstrong RA, Cruz Motta JJ, **Courtney TA**. 2025. Hypoxia driven by Caribbean Sargassum accumulation events. Royal Society Open Science 12(8):250322. https://doi.org/10.1098/rsos.250322
35. **Courtney TA** & Andersson AJ. 2025. Revisiting the terms used in net ecosystem calcification studies, carbonate budgets, and the equation of coral reef growth. Coral Reefs 44: 1031–1036. https://doi.org/10.1007/s00338-025-02643-8
34. Sanassy Pilly S, Townsend JE, Alisa CA, Razak TB, Roche RC, Turner JR, Chan S, Kriegman DJ, Andersson AJ, Perry CT, Lange ID, **Courtney TA**. 2025. Quantifying coral reef carbonate budgets: a comparison between *ReefBudget* and *CoralNet*. Coral Reefs 44: 513–527. https://doi.org/10.1007/s00338-025-02620-1
33. Pezner AK, **Courtney TA**, Chou W-C, Chu H-C, Frable BW, Kekuewa SAH, Soong K, Wei Y, Andersson AJ. 2024. Coral growth along a natural gradient of seawater temperature, pH, and oxygen in a nearshore

seagrass bed on Dongsha Atoll, Taiwan. PLoS ONE 19(10): e0312263. https://doi.org/10.1371/journal.pone.0312263

32. Mejias-Rivera CL & **Courtney TA**. 2024. Ocean Warming, Heat Stress, and Coral Bleaching in Puerto Rico. Caribbean Journal of Science (54): 132-149. https://doi.org/10.18475/cjos.v54i1.a17
31. Toth LT, **Courtney TA**, Colella M, Ruzicka RR. 2023. Stony coral tissue loss disease accelerated shifts in coral composition and declines in reef accretion potential in the Florida Keys. Frontiers in Marine Science. https://doi.org/10.3389/fmars.2023.1276400
30. Mejias-Rivera CL, Armstrong RA, Balint S, García-Troche E, McKinney RA, Morell JM, Oczkowski A, **Courtney TA**. 2023. Localized inshore warming, acidification, and elevated particulate organic matter across a coupled mangrove, seagrass, and coral reef ecosystem in La Parguera, Puerto Rico. Coral Reefs. https://doi.org/10.1007/s00338-023-02435-y
29. Cornwall CE, Carlot J, Branson O, **Courtney TA**, Harvey BP, Perry CT, Andersson AJ, Diaz-Pulido G, Johnson MD, Kennedy E, Krieger EC, Mallela J, McCoy SJ, Nugues MM, Quinter E, Ross CL, Ryan E, Saderne V, Comeau S. 2023. Crustose coralline algae can contribute more than corals to coral reef carbonate production. Nature Communications Earth & Environment. https://doi.org/10.1038/s43247-023-00766-w
28. Pezner AK, **Courtney TA**, Barkley HC, Chou WC, Chu HC, Clements SM, Cyronak T, DeGrandpre MD, Kekuewa SAH, Kline DI, Liang Y-B, Martz TB, Mitarai S, Page HN, Rintoul MS, Smith JE, Soong K, Takeshita Y, Tresguerres M, Wei Y, Yates KK, Andersson AJ. 2023. Increasing hypoxia on global coral reefs under ocean warming. Nature Climate Change 13, 403–409 https://doi.org/10.1038/s41558-023-01619-2
27. Rintoul M, **Courtney T**, Dohner J, Giddings S, Inoha K, Kekuewa S, Mitarai S, Monismith S, Pezner A, Andersson A. 2022. The Effects of Light Intensity and Flow Speed on Biogeochemical Variability within a Fringing Coral Reef in Onna-son, Okinawa, Japan. Journal of Geophysical Research – Oceans. https://doi.org/10.1029/2021JC018369
26. Kekuewa SAH, **Courtney TA**, Cyronak T, Andersson AJ. 2022. Seasonal nearshore ocean acidification and deoxygenation in the Southern California Bight. Scientific Reports. https://doi.org/10.1038/s41598-022-21831-y
25. Toth LT, **Courtney TA**, Colella MA, Kupfner Johnson SA, Ruzicka RR. 2022. The past, present, and future of coral reef growth in the Florida Keys. Global Change Biology. https://doi.org/10.1111/gcb.16295
24. **Courtney TA**, Barkley HC, Chan S, Couch CS, Kindinger TL, Oliver TA, Kriegman DJ, Andersson AJ. 2022. Rapid assessments of Pacific Ocean net coral reef carbonate budgets and net calcification following the 2014-2017 global coral bleaching event. Limnology and Oceanography. https://doi.org/10.1002/lno.12159
23. Elahi R, Edmunds PJ, Gates RD, Kuffner IB, Barnes BB, Chollett I, **Courtney TA**, Guest JR, Lenz EA, Toth LT, Viehman TS, Williams ID. 2022. Scale dependence of coral reef oases and their environmental correlates. Ecological Applications. https://doi.org/10.1002/eap.2651
22. Bresnahan P, Cyronak T, Brewin RJW, Andersson A, Wirth T, Martz T, **Courtney T**, Hui N, Kastner R, Stern A, McGrain T, Reinicke D, Richard J, Hammond K, Waters S. 2022. A High-Tech, Low-Cost, Internet of Things Surfboard Fin for Coastal Citizen Science, Outreach, and Education. Continental Shelf Research. https://doi.org/10.1016/j.csr.2022.104748
21. **Courtney TA**, Cyronak T, Griffin AJ, Andersson AJ. 2021. Implications of salinity normalization of seawater total alkalinity in coral reef metabolism studies. PLOS ONE. https://doi.org/10.1371/journal.pone.0261210
20. Kekuewa SAH, **Courtney TA**, Cyronak T, Kindeberg T, Eyre BD, Stoltenberg L, Andersson AJ. 2021. Temporal and Spatial Variabilities of Chemical and Physical Parameters on the Heron Island Coral Reef Platform. Aquatic Geochemistry. https://doi.org/10.1007/s10498-021-09400-7
19. **Courtney TA**, Guest JR, Edwards AJ, Dizon RM. 2021. Linear extension, skeletal density, and calcification rates of the blue coral *Heliopora coerulea*. Coral Reefs. https://doi.org/10.1007/s00338-021-02137-3
18. Pezner AK, **Courtney TA**, Page HN, Giddings SN, Beatty CM, DeGrandpre MD, Andersson AJ. 2021. Lateral, Vertical, and Temporal Variability of Seawater Carbonate Chemistry at Hog Reef, Bermuda. Frontiers in Marine Science. https://doi.org/10.3389/fmars.2021.562267

17. **Courtney TA**, Kindeberg T, Andersson AJ. 2020. Coral calcification responses to the North Atlantic Oscillation and coral bleaching in Bermuda. PLOS ONE. https://doi.org/10.1371/journal.pone.0241854
16. Kindeberg T, Bates NR, **Courtney TA**, Cyronak T, Griffin A, Mackenzie FT, Paulsen M-L, Andersson AJ. 2020. Porewater Carbonate Chemistry Dynamics in a Temperate and a Subtropical Seagrass System. Aquatic Geochemistry. https://doi.org/10.1007/s10498-020-09378-8
15. **Courtney TA**, Barnes BB, Chollett I, Elahi R, Gross K, Guest JR, Kuffner IB, Lenz EA, Nelson HR, Rogers CS, Toth LT, Andersson AJ. 2020. Disturbances drive changes in coral community assemblages and coral calcification capacity. Ecosphere. https://doi.org/10.1002/ecs2.3066
14. Cyronak T, Takeshita Y, **Courtney TA**, DeCarlo EH, Eyre BD, Kline DI, Martz T, Page H, Price NN, Smith J, Stoltenberg L, Tresguerres M, Andersson AJ. 2020. Diel temperature and pH variability scale with depth across diverse coral reef habitats. Limnology and Oceanography Letters. https://doi.org/10.1002/lol2.10129
13. Baumann JH, Ries JB, Rippe JP, **Courtney TA**, Aichelman HE, Westfield I, Castillo KD. 2019. Nearshore coral growth declining on the Mesoamerican Barrier Reef System. Global Change Biology, 25: 3932–3945. https://doi.org/10.1111/gcb.14784
12. **Courtney TA** & Andersson AJ. 2019. Evaluating measurements of coral reef net ecosystem calcification rates. Coral Reefs, 38(5):997–1006. https://doi.org/10.1007/s00338-019-01828-2
11. Page HN, **Courtney TA**, De Carlo EH, Howins NM, Koester I, Andersson AJ. 2019. Spatiotemporal variability in seawater carbon chemistry for a coral reef flat in Kāne'ohe Bay, Hawai'i. Limnology and Oceanography, 63:913–934. https://doi.org/10.1002/lno.11084
10. Guest JR, Edmunds PJ, Gates RD, Kuffner IB, Andersson AJ, Barnes BB, Chollett I, **Courtney TA**, Elahi R, Gross K, Lenz EA, Mitarai S, Mumby PJ, Nelson HR, Parker BA, Putnam HM, Rogers CS, Toth LT. 2018. A framework for identifying and characterising coral reef "oases" against a backdrop of degradation. Journal of Applied Ecology, 00:1-11. https://doi.org/10.1111/1365-2664.13179
9. **Courtney TA**, De Carlo EH, Page HN, Bahr KD, Barro A, Howins N, Tabata R, Terlouw G, Rodgers KS, Andersson AJ. 2018. Recovery of reef-scale calcification following a bleaching event in Kāne'ohe Bay, Hawai'i. Limnology & Oceanography Letters, 3:1–9. https://doi.org/10.1002/lol2.10056
8. **Courtney TA**, Lebrato M, Bates NR, Collins A, de Putron SJ, Garley R, Johnson, R, Molinero JC, Noyes TJ, Sabine CL, Andersson AJ. 2017. Environmental controls on modern scleractinian coral and reef-scale calcification. Science Advances, 3(11), p.e1701356. https://doi.org/10.1126/sciadv.1701356
7. Page HN, **Courtney TA**, Collins A, De Carlo EH, Andersson AJ. 2017. Net community metabolism and seawater carbonate chemistry scale non-intuitively with coral cover. Frontiers in Marine Science, 4:161. https://doi.org/10.3389/fmars.2017.00161
6. **Courtney TA**, Andersson AJ, Bates NB, Collins A, Cyronak T, de Putron SJ, Eyre BD, Garley R, Hochberg EJ, Johnson R, Musielewicz S, Noyes T, Sabine CL, Sutton AJ, Toncin J, Tribollet A. 2016. Comparing Chemistry and Census-based Estimates of Net Ecosystem Calcification on a Rim Reef in Bermuda. Frontiers in Marine Science, 3:181. https://doi.org/10.3389/fmars.2016.00181
5. Baumann JH, Townsend JE, **Courtney TA**, Aichelman HE, Davies SW, Lima FP, Castillo KD. 2016. Temperature Regimes Impact Coral Assemblages along Environmental Gradients on Lagoonal Reefs in Belize. PLoS ONE, 11(9): e0162098. https://doi.org/10.1371/journal.pone.0162098
4. Aichelman HE, Townsend JE, **Courtney TA**, Baumann JH, Davies SW, Castillo KD. 2016. Heterotrophy mitigates the response of the temperate coral *Oculina arbuscula* to temperature stress. Ecology and Evolution, 6(18): 6758-6769. https://doi.org/10.1002/ece3.2399
3. Horvath KM, Castillo KD, Armstrong P, Westfield IT, **Courtney T**, Ries JB. 2016. Next-century ocean acidification and warming both reduce calcification rate, but only acidification alters skeletal morphology of reef-building coral *Siderastrea siderea*. Scientific Reports, 6:29613. https://doi.org/10.1038/srep29613
2. **Courtney T** and Ries JB. 2015. Impact of atmospheric pCO_2, seawater temperature, and calcification rate on the $\delta^{18}O$ and $\delta^{13}C$ composition of echinoid calcite (*Echinometra viridis*). Chemical Geology, 411: 228-239. https://doi.org/10.1016/j.chemgeo.2015.06.030

1. **Courtney T**, Ries JB, Westfield I. 2013. Predicted end of 21st century CO_2-induced ocean acidification decreases calcification rates in the tropical urchin *Echinometra viridis*. Journal of Experimental Marine Biology and Ecology, 440: 169-175. https://doi.org/10.1016/j.jembe.2012.11.013

Computational tools and datasets:

3. **Courtney TA,** Lange ID, Sannassy Pilly, S, Townsend JE, Chan S, Perry CT, Kriegman DJ, Andersson AJ. 2024. Area-normalized scaling of ReefBudget calcification, macrobioerosion, and microbioerosion rates for use with CoralNet Version 2.0. Zenodo. https://doi.org/10.5281/zenodo.13257132
3. **Courtney TA**, Andersson AJ. 2021. Calcification Dissolution Potential Tool for Excel: Version 1. Zenodo. https://doi.org/10.5281/zenodo.7051628
2. Chan S, **Courtney TA**, Andersson AJ, Kriegman DJ. 30 July 2021. "CoralNet now estimates carbonate production rates." CoralNet. https://coralnet.ucsd.edu/blog/coralnet-now-estimates-carbonate-production-rates/
1. **Courtney TA**, Chan S, Lange ID, Perry CT, Kriegman DJ, Andersson AJ. 2021. Area-normalized scaling of ReefBudget calcification, macrobioerosion, and microbioerosion rates for use with CoralNet Version 1.0. Zenodo. https://doi.org/10.5281/zenodo.5140477

Conference presentations and invited talks:

28. **Courtney TA**. 2026. Developing and Validating Automated Tools to Assess Coral Reef Carbonate Production. University of Puerto Rico Mayagüez Department of Geology Annual Symposium.
27. **Courtney TA**. 2025. Coral bleaching and coastal hypoxia in Puerto Rico. University of Puerto Rico Mayagüez Department of Biology Seminar invited speaker.
26. **Courtney TA** & Cruz Motta JJ 2025. Monitoring water quality, fish and corals in Puerto Rico's coral reefs and applications of the Biological Condition Gradient (BCG). Puerto Rico Department of Natural and Environmental Resources.
25. **Courtney TA**. 2023. Coral reef calcification and biogeochemistry under environmental change. University of Puerto Rico Mayagüez Department of Chemistry Seminar invited speaker.
24. **Courtney TA**. 2022. Blue carbon uptake and buffering against ocean acidification by seagrasses. EcoEléctrica invited talk.
19. **Courtney TA**. 2022. Benthic metabolism and buffering against ocean acidification. University of Puerto Rico Mayagüez – Purdue University Blue Initiative Partnership Meeting.
23. **Guest lecture**. 2021. Ocean Briefs Feedback and Discussion for University of Montana Story Lab
22. **Courtney TA** & Andersson AJ. 2021. Tools to assess coral reef calcification. NOAA Ocean Acidification Working Group Meeting.
21. **Courtney TA**. 2021. Coral reef calcification under rapid environmental change. AECiMA Outreach Talks: University of Puerto Rico, Mayagüez.
20. **Courtney TA**. 2021. Coral reef metabolism and biogeochemistry under rapid environmental change. University of Puerto Rico, Mayagüez.
19. **Guest lecture**. 2020. "Disturbances and coral reef calcification" for NSF Summer REU Program
18. **Guest lecture**. 2020. Rapid estimates of coral reef calcification for SIO 119 Undergraduate Course
17. **Guest lecture**. 2020. "The chemistry (and physics) of quantifying net coral reef calcification" for SIO 119
16. **Courtney TA**. 2020. Disturbances drive changes in coral community assemblages and coral calcification capacity (and the development of tools to assess reef-scale calcification under OA). NOAA Ocean Acidification Community Meeting.
15. **Courtney TA**. 2019. Coral calcification and climate change. Scripps Education Association.
14. **Courtney TA** and Andersson AJ. 2019. Evaluating measurements of ecosystem-scale coral reef calcification under global environmental change. ASLO Aquatic Sciences Meeting.
13. **Guest lecture.** 2019. "Marine Chemistry Basics" for SIO Master of Advanced Studies
12. **Guest lecture.** 2019. Paleoclimatology Lab for SIO Master of Advanced Studies (with Dr. Art Miller)

11. **Courtney TA**. 2018. The science (and value) of coral reef calcification. San Diego State Environmental Business Society.
10. **Courtney TA**. 2018. Environmental controls on coral and reef-scale calcification. National Sun Yat-sen University lunch seminar.
9. **Courtney TA**, Lebrato M, Bates NR, Collins A, de Putron SJ, Garley R, Johnson, R, Molinero JC, Noyes TJ, Sabine CL, Andersson AJ. 2018. New insights into the drivers of coral and reef-scale calcification from Bermuda. Ocean Sciences Meeting.
8. **Courtney TA**. 2017. Environmental controls on coral and reef-scale calcification. San Diego Coral Club.
7. **Courtney TA**, Andersson AJ, De Carlo EH, Page HN, Koester I, Terlouw G, Tabata R, Bahr KD, Rodgers KS. 2017. Coral bleaching impacts on reef-scale net calcification and net community production in Kāne'ohe Bay, HI. ASLO Ocean Sciences Meeting.
6. **Courtney TA** and Andersson AJ. 2016. Seasonal patterns in calcium carbonate production of a Bermuda coral reef. Scripps Student Symposium.
5. **Courtney TA**, Andersson AJ, Cyronak T, Noyes T, Bates NR, Collins A, de Putron S, Garley R, Hochberg EJ, Johnson R. 2016. Comparing chemistry and census-based estimates of net ecosystem calcification on a rim reef in Bermuda. 13th International Coral Reef Symposium.
4. **Courtney T**, Baumann J, Foguel AD, Horvath K, Westfield I, Castillo KD, Ries JB. 2014. Characterizing 21st century growth trends of the scleractinian coral *Siderastrea siderea* throughout the Belize barrier reef and atoll system. Benthic Ecology Meeting.
3. **Courtney T**, Ries JB, Westfield I. 2013. Impact of atmospheric pCO_2 and seawater temperature on calcification rate and stable isotopic composition ($\delta^{18}O$, $\delta^{13}C$) of echinoid calcite (*Echinometra viridis*). UNC at Chapel Hill Anadarko Research Symposium.
2. **Courtney T**, Ries JB, Westfield I. 2012. *Echinometra viridis* exhibits seasonal response in calcification rates to predicted end of 21st century CO_2-induced ocean acidification. Geological Society of America Fall Meeting.
1. **Courtney T**, Ries JB, Westfield I. 2012. Effects of warming and CO_2-induced acidification on the tropical urchin *Echinometra viridis*. UNC at Chapel Hill Department of Marine Sciences Seminar Series.

Fieldwork experience:

2025 **La Parguera, Puerto Rico**: Monthly seawater chemistry sampling
2024 **La Parguera, Puerto Rico**: Coral reef benthic surveys, instrument deployments
2023 **Hawai'i**: E/V Nautilus Cruise NA 156 - Ocean Exploration Through Advanced Imagery
2023 **La Parguera, Puerto Rico**: Seagrass chemistry surveys, instrument deployments, cores, and productivity
2019 **Okinawa, Japan**: Chemistry surveys, instrument deployments, and lab experiments
2019 **Taiping Island, Taiwan (lead)**: Chemistry surveys, instrument deployments, and benthic surveys
2018 **Bermuda (lead)**: MAPCO2 buoy servicing
2018 **Dongsha Atoll, Taiwan**: Chemistry surveys, porewaters, instrument deployments, and coral coring
2017 **Bermuda**: MAPCO2 buoy servicing, seawater chemistry surveys, and instrument deployments
2017–2021 **La Jolla Bight, California:** Chemistry surveys and instrument deployments
2017–2018 **Mission Bay, California:** Chemistry surveys, porewaters, and instrument deployments
2017 **Kāne'ohe Bay, Hawai'i (lead)**: Chemistry surveys
2016 **Kāne'ohe Bay, Hawai'i**: Chemistry surveys, porewaters, and instrument deployments
2016 **Bermuda**: MAPCO2 buoy servicing and coral coring
2016 **Kāne'ohe Bay, Hawai'i**: Chemistry surveys, benthic surveys, and mesocosm experiments
2014 **Belize Barrier Reef System**: Coral reef community benthic surveys
2014 **Southern Outer Banks, North Carolina**: Coral sample collection for lab experiments
2013 **Southern Outer Banks, North Carolina**: Benthic habitat surveys
2012 **Belize Barrier Reef System**: Coral coring

Teaching experience:

2026 Spring **Instructor:** Chemical Oceanography, University of Puerto Rico Mayagüez

2025 Fall **Instructor:** Open Data Science in Chemical Oceanography, University of Puerto Rico Mayagüez
2025 Spring **Instructor:** Chemical Oceanography, University of Puerto Rico Mayagüez
2024 Fall **Instructor:** Marine Pollution, University of Puerto Rico Mayagüez
2024 Spring **Instructor:** Chemical Oceanography, University of Puerto Rico Mayagüez
2023 Fall **Instructor:** Marine Pollution, University of Puerto Rico Mayagüez
2023 Spring **Instructor:** Chemical Oceanography, University of Puerto Rico Mayagüez
2023 Spring **Instructor:** Chemical Oceanography Data Analysis Laboratory, University of Puerto Rico Mayagüez
2022 Fall **Instructor:** Marine Pollution, University of Puerto Rico Mayagüez
2022 Spring **Instructor:** Chemical Oceanography, University of Puerto Rico Mayagüez
2022 Spring **Instructor:** Chemical Oceanography Data Analysis Laboratory, University of Puerto Rico Mayagüez
2021 Fall **Instructor:** Marine Pollution, University of Puerto Rico Mayagüez
2021 Fall **Instructor:** Professional Ethics in Marine Sciences, University of Puerto Rico Mayagüez
2018 **Instructor**: SIO 90 Perspectives on Ocean Science
2013 **Teaching Assistant**: GEOL 101L Introductory Geology Lab, UNC

Graduate students mentored:

University of Puerto Rico at Mayagüez Committee Chair:

2025–present **Kenneth Montalvo Negron,** MS, Scaling up seaweed production in the Caribbean
2025–present **Amanda Melendez Perez,** MS, Metabolism and carbonate production of soft corals
2025–present **Alejandra Aguilar-Giraldo,** PhD, Quantifying rates and drivers of coral reef bioerosion
2023–present **Joseph Townsend,** PhD, Rates, drivers, and restoration of coral reef carbonate production
2024–2026 **Janluis Rivera,** MS, Modeling wave attenuation by coral restoration (co-chair: Dr. Andre Amador)
2023–2026 **Carolina Melendez Declet,** MS, Designing integrated multitrophic aquaculture for *Ulva* seaweed
2023–2025 **Ignacio Rueda,** MS, Photogrammetry to quantify structural complexity provisioning by deep-sea corals
2022–2025 **Juanita Carballeira Martinez,** MS, Vertical, spatial, and temporal variability in coral reef biogeochemistry
2022–2025 **Carla Mejías Rivera,** PhD, Remote sensing and in situ assessments of water quality in Puerto Rico
2022–2025 **Irais Luquis Ramos,** MS, Modeling *Diadema* sea urchin bioerosion to inform restoration strategies
2022–2024 **Jose Martinez Ortiz,** MS, Sargassum accumulation events as drivers of Caribbean coastal hypoxia
2022–2024 **Catherine Hernández Rodriguez,** MS, Seagrass carbon cycling, pH buffering, and carbon sequestration
2021–2023 **Leira Centeno Mejias**, MS, Water quality monitoring compliance project

Scripps Institution of Oceanography Informal Mentoring:

2019–2021 **Ariel Pezner**, PhD, Impacts of hypoxia on coral reefs
2019–2021 **Sam Kekuewa**, PhD, Spatiotemporal variability of seawater chemistry in coastal ecosystems
2018–2019 **Thompson Banez**, MAS in Marine Biodiversity and Conservation Student Mentor
2017–2018 **Emily Parker**, MAS in Marine Biodiversity and Conservation Student Mentor
2017–2018 **Sam Kekuewa**, MS Graduate Peer Mentor Program
2017–2018 **Wiley Wolfe**, PhD Graduate Peer Mentor Program

Undergraduate students mentored:

2026-present **Sediment characterization relative to coral reef carbonate production (UPRM):** Ariana Negron Pacheco
2025-2026 **Segmentation of large-area coral reef imagery (UPRM)**: Paola Berrios Morales, Ariana Negron Pacheco
2023-2024 **Analysis of seawater quality (UPRM)**: Victoria Reyes Claudio, Claudia Lebron Moldonado
2023 **Blue Integrated Partnerships Summer Internship Program (UPRM)**:
Alexis Warnecke & Victoria Reyes Claudio *Awarded 1st Place for Best Presentation
2020–2021 **Seawater carbonate chemistry and inshore-offshore gradients of coral reef biogeochemistry (UCSD)**:
William Tallentire
Summer 2020 **Analysis of coral cover following coral bleaching, NSF Research Experiences for Undergraduates (UCSD):**
Audrey Ellias Zach Ferris
2013–2014 **Coral skeletal geochemistry, growth rate analysis, and marine aquarium facilities (UNC-CH)**:
Hannah Aichelman Courtney Anderson Madelyn Roycroft

Pualani Armstrong	Jessica Boulton	Joseph Townsend
Carissa Campbell	Kathryn Cobleigh	Vallari Eastman
Ashley Foguel	Hannah Knight	

Graduate student committees:

2026–present **Elianis Quintana Abrams**, UPRM PhD in Marine Sciences
2026–present **Angel Acosta Colon**, UPRM PhD in Marine Sciences
2026–present **Kevin Martinez Lopez**, UPRM MS in Marine Sciences
2026–present **Mariela Cortes Medina**, UPRM MS in Marine Sciences
2026–present **Carolina Cesar Avila**, UPRM MS in Marine Sciences
2025–present **Jomar Camacho Garay,** UPRM MS in Marine Sciences
2025–present **Melanie Cordero Cruz**, UPRM MS in Marine Sciences
2023–present **Jenniffer Perez Perez**, UPRM PhD in Marine Sciences
2023–present **Adiana Bayo Torres**, UPRM MS in Marine Sciences
2023–present **Maria Jose Brito Vera**, UPRM PhD in Marine Sciences
2025–2026 **Glorimar Franqui Rivera**, UPRM PhD in Marine Sciences
2024–2026 **Andrea Velez Rivera**, UPRM MS in Marine Sciences
2024–2026 **Biajani Gonzalez Torres**, UPRM MS in Marine Sciences
2022–2025 **Raymond Infante Rosa**, UPRM MS in Marine Sciences
2022–2024 **Ana Medina Martinez**, UPRM MS in Marine Sciences

Working groups, workshops, and field courses:

2024 & 2025 **Drivers of ecological stability in response to climate change across ecosystems**, USGS Powell Center
2024 **Puerto Rico Coral Reef Restoration Action Plan**, ISER Caribe, NOAA, DRNA, and The Nature Conservancy
2023 **Caribbean Carbon Accounting in Seagrass**, Florida International University and The Nature Conservancy
2023 **D-ENTERPRISE Conference to Increase Diversity in the Ocean Sciences**, Office of Naval Research
2022 **Will coral reefs become built by calcifying seaweed in the future?**, French Embassy in New Zealand
2017 **Local-scale coral reef resilience under global-scale ocean change**, USGS Powell Center
2017 **Coral In Situ MEtabolism (CISME) workshop**, Kāne'ohe Bay, Hawai'i Institute of Marine Biology
2016 **Natural History Below the Tides,** La Jolla, Scripps Institution of Oceanography
2012 **Field Geology in the Rocky Mountains**, Montana, Judson Mead Geologic Field Station
2011 **Human and Marine Ecology,** Galápagos Islands, Universidad San Francisco de Quito
2011 **Marine Resources Population Dynamics Workshop**, NOAA NMFS
2010 **Alternative Fall Break Environmental Trip Co-Leader**: UNC APPLES Service Learning

Outreach and service:

2024 Panelist on NSF GRFP for UNC Chapel Hill Coral Reefs and Climate Change course
2023 Ocean Exploration with Dr. Bob Ballard Livestream for University of Puerto Rico
2020 SciREN scientist participant, created lesson plan on coral reef growth for K-12 teachers
2020–2021 Mentor for Científico Latino Graduate Student Mentorship Initiative
2019 University of California delegate to the 25th United Nations Conference of Parties
2019–2021 Letters to a Pre-Scientist participant
2018 Panelist for San Diego State University Earth Week Chasing Coral screening
2018 Ocean Acidification lecture for UCSD Retirement Association
2018 Panelist for UCSD Retirement Association Chasing Coral screening
2017 Panelist for Citizens Climate Lobby Chasing Coral screening
2017 Panelist for Smartfin + Changing Tides Chasing Coral screening
2015–2021 Smartfin project surfboard fin sensors ß–tester
2015–2020 Scripps Community Outreach Program for Education (SCOPE) volunteer
2015–2016 Rosa Parks Elementary School volunteer tutor

2015, 16, 18 Ocean Discovery Institute volunteer scientist
2014 SciREN scientist participant, created lesson plan on ocean acidification for K-12 teachers
2012 NC Museum Natural Sciences Marine Mammal Day volunteer

Job related certifications:

2021 Virtual Teacher Certificate, University of California Irvine
2020–present Freediving Instructors International Level 1 Freediver
2020 Introduction to College Teaching, University of California San Diego
2018–present California Boater License
2017–present Motorboat Operator Certification Course
2011–present AAUS Science Diver to 60 ft. + nitrox (>200 dives)
2011–present First Aid, CPR, Emergency O_2, and Diving Neurological Assessment Certified
2010–present North Carolina Boater Education Certified

Diversity, Equity, and Inclusion Trainings

2023 Harassment and Discrimination Prevention for Non-Supervisors, EasyLlama
2020 Introduction to College Teaching, UC San Diego Teaching + Learning Commons
2020 Self-Guided Foundational Safe Zone Training, Safe Zone Project
2020 Transfer Ally Training, UC San Diego
2020 Undocu-Ally Training, UC San Diego
2020 Conflict de-escalation Training, Hollaback!

Academic service:

2025–2026 UPRM Department of Marine Sciences – Curriculum Committee Member
2024–Present UPRM College of Arts and Sciences – Dialogue Committee Member
2022–Present UPRM Department of Marine Sciences – Graduate Committee Chair
2021–Present UPRM Department of Marine Sciences – Appraisal Committee Member
2021–2024 UPRM Department of Marine Sciences – Diving Control Board Member
2022 National Science Foundation Review Panelist
2021 Sea Grant Puerto Rico Technical Review Panelist
2019 Session co-chair: Coral Reef Ecosystems, ASLO Aquatic Sciences Meeting
2017–2019 Treasurer + Operations, Scripps Academic Club

Peer review:

Global Change Biology
Hydrobiologia
Proceedings of the Royal Society B
Marine Environmental Research
Geophysical Research Letters
National Science Foundation
Marine Ecology Progress Series
Hawai'i Sea Grant
Communications Earth & Environment
Palaeogeography, Palaeoclimatology, Palaeoecology
Journal of Experimental Marine Biology and Ecology
Regional Studies in Marine Science
Science of the Total Environment
Ecosphere
Marine & Freshwater Research
Earth Surface Processes and Landforms
Nature Climate Change
Limnology and Oceanography
Coral Reefs
Bulletin of Marine Science
PLOS ONE
Marine Chemistry
One Earth
Marine Ecology
Scientific Reports
PeerJ
Frontiers in Marine Science
Limnology and Oceanography: Methods
PNAS
Global Biogeochemical Cycles
Limnology and Oceanography Letters